MILITARY SPECIFICATION
CHARGE, PROPELLING ASSEMBLY (SINGLE BAG) 105MM, HOWITZER M137 (MOD)

1. SCOPE

1.1 This specification covers the loading assembling and packing for one type of propellant charge for use in Cartridge 105MM, HE, M1 (MOD) or WP, M60 (MOD) for 105MM Howitzer M137.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-843 -Cushioning Material, Cellulosic

MILITARY

MIL-A-2550 -Ammunition and Special Weapons, General Specification for
MIL-C-45662 -Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 -Quality Assurance Terms and Definitions
MIL-STD-1168-Lot Numbering of Ammunition
MIL-STD-1235-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.
MS-63053 -Drum, Metal Shipping and Storage 15 and 19 Gallon

DRAWINGS

9217087 -Marking Diagram and Sealing of Steel Drums for Shipment of Propellants, Bagged Charges
9276669 -Charge, Propelling Assembly

FSC: 1315

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling Charge Assembly.-The propelling charge assembly shall comply with all requirements specified on Drawing (dwg) 9276669 and with all requirements specified in applicable specifications.

3.3 Weight.-The net weight of the propellant shall not differ by more than 0.02 ounce from the amount specified in the loading authorization.

3.4 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. All parts shall be free of dirt, grease, and other foreign material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

d. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents. and the certificates of conformance are on file and available for review.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All materials submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain propellant from not more than one lot.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable

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subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Bag, Propellant (see dwg. 9276668 covering a detail of dwg. 9276669).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	None defined	
Minor:	AQL 1.5 percent	
201.	Any seam or opening incompletely stitched	Visual
202.	Marking missing misleading or unidentifiable	Visual
203.	Foil missing or improperly assembled	Visual
204.	Evidence of poor workmanship (see 3.4)	Visual

4.2.2.2 Charge, Propelling (see dwg. 9276669).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Assembly damaged to the extent that propellant can escape	Visual
Minor:	AQL 1.5 percent	
201.	Any seam or opening incompletely stitched	Visual
202.	Thread broken	Visual
203.	Evidence of poor workmanship (see 3.4)	Visual

4.2.2.3 Drum, prior to filling (see dwg. MS63053).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Foreign material in drum	Visual
Minor:	None defined	

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4.2.2.4 Drum sealed see dwg. MS63053 and 9217087).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Gasket missing or improperly positioned...	Visual
102.	Locking ring improperly positioned.....	Visual
Minor:	None defined	

4.2.2.5 Fiber Container

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	AQL 1.0 percent	
101.	Gasket missing or incomplete.....	Visual
102.	Locking device damaged or improperly closed.....	Visual
103.	Holes or breaks in cover or body.....	Visual
104.	Damage to coating or cover.....	Visual
105.	Foreign matter or corrosion in interior...	Visual
Minor:	AQL 1.0 percent	
201.	Marking missing, misleading or unidentifiable.....	Visual
202.	Interior or exterior torn or delaminated..	Visual

4.2.3 Testing

4.2.3.1 Weight-(See 3.3) Major defect-The propellant charge shall be weighed and checkweighed 100 percent, any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. (CD)

4.2.3.1.1 Method 1-The propellant shall be weighed in a container and placed in the bag, the bag sewn and the propellant charge assembly weighed. The net weight of the propellant may be calculated using an average tare weight of the empty bags or the tare weight as marked on the individual bag.

4.2.3.1.2 Method 2-The propellant in a container shall be weighed and then checked weighed on a different balance and if performed manually, by an other operator. The propellant shall then be placed in the bag and the bag sewn. The weight of the loaded bags should be checked as a major defect using

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a sampling plan from MIL-STD-105 or MIL-STD-1235 with an AQL of 0.40 percent. The net weight of the propellant shall be calculated using an average weight of the empty bag or the tare weight as marked on the individual bag.

4.2.3.2 Air pressure test.(see 4.3.1) Major defect- The containers shall be checked 100 percent as specified in 4.3.1. If any container fails to meet the requirement it shall be rejected and removed from the lot. (CD)

4.2.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (see 6.3).

4.2.4.1 Inspection Equipment Capability and Accuracy.-All inspection equipment and facilities shall be of adequate quality and quantity to perform properly and accurately all the required inspections, measurements and tests. The inspection equipment shall be maintained in satisfactory condition for the continued proper performance of these functions and shall be calibrated in accordance with the provisions of MIL-C-45662.

4.2.4.2 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3 Test Methods and Procedures

4.3.1 Air pressure test.-The container shall be subject to an internal pressure of 2 to 3 pounds per square inch (P.S.I.) by a method satisfactory to the contracting officer's representative. A gage (designed to read 1/4 P.S.I) or a water manometer shall be assembled in the system. A drop of 1/4 P.S.I. on the gage or 0.7 inches on the manometer in minute shall be cause for rejection. (Applicable to steel drums only).

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The propelling charges shall be packed in containers conforming to Standard MS 63053-2. The gross weight of the pack shall not exceed 150 pounds. Cushioning material conforming to PPP-C-843 shall be added as required to provide a tight pack.

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5.1.2 Level C.-The charges shall be packed in a fiber drum certified in accordance with ICC Specification 21C and meeting the following requirements. Size shall be 15-1/2 ± 1/2 inches in diameter by 26 ± 1 inches deep. A layer of aluminum foil shall be laminated into the body and the bottom if it is made of fiberboard. The bottom shall be either steel or asphalt laminated fiberboard. The bottom and top edge of the drum shall have 24 gauge welded galvanized steel chimes. The cover shall be either plain or interlocking (with bottom) design, lever lock type with gasket, and shall provide a moisture-proof seal. The cover shall be made with either a galvanized finish or a baked on varnish finish resistant to chipping, peeling and cracking. The drum shall be tested in accordance with ICC Specification 21C except the drum shall be filled with not less than 250 pounds of dry sand for the drop test. The exterior of the drum shall be coated with a water-resistant material.

6. NOTES

6.1 Ordering data.-Procurement documents should specify title, number and date of this specification.

6.2 Intended use.-The charge covered by this specification is intended for use with fixed 105MM Howitzer ammunition for Air Force use only.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

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6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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